

Project Document: NYC Taxi Trip Duration Prediction

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1. Project Objective

The primary objective of this mini-project is to develop a machine learning model capable of predicting NYC taxi trip durations. Beyond basic model development, a significant focus is placed on demonstrating key MLOps practices, including MLflow for experiment tracking, model versioning, data drift simulation, and model serving through a self-contained Docker image. The goal is to build a robust and maintainable ML pipeline.

2. Data Loading and Initial Exploration

Process:

- The raw taxi trip data (NY_Trip_Raw_Data.csv) was loaded into a pandas DataFrame.
- Initial exploration involved using `df.info()` to inspect data types and non-null counts, `df.isnull().sum()` to identify missing values, and `df.describe()` to get descriptive statistics of numerical columns.

Observations:

- The dataset contains approximately 627,097 entries and 12 columns, including trip ID, vendor ID, pickup/dropoff datetimes, passenger count, weather conditions, GPS coordinates, store-and-forward flag, and trip duration.
- A small number of missing values (1 in most columns, 4 for GPS coordinates due to a single row) were identified.
- `pickup_datetime` and `dropoff_datetime` were initially loaded as objects (strings) and required conversion.

- `trip_duration` is the target variable, ranging from 1 second to over 1.9 million seconds, suggesting potential outliers.

3. Data Preprocessing and Feature Engineering

Process:

- **Schema Validation:**
 - `pickup_datetime` and `dropoff_datetime` columns were converted to datetime objects, with errors='coerce' to handle unparseable entries (none found).
 - Rows where `dropoff_datetime` was before `pickup_datetime` were removed (none found).
 - Rows with missing GPS pings (4 rows) were identified.
 - Rows with any invalid timestamps were dropped.
- **Feature Engineering:**
 - **`pickup_hour`:** Extracted the hour from `pickup_datetime`.
 - **`pickup_dayofweek`:** Extracted the day of the week (Monday=0, Sunday=6) from `pickup_datetime`.
 - **`is_weekend`:** A binary feature indicating if the `pickup_dayofweek` is Saturday or Sunday.
 - **`distance_km`:** Calculated the Haversine distance between pickup and dropoff coordinates.
 - **`weather`:** Converted to a categorical data type.
- **Dataset Versioning:** The processed data was saved to a new CSV file (`NY_Trip_Processed_Data_v1.csv`) for version control and reproducibility.

4. Model Training and Comparison

Process:

- **Data Preparation:** The `id`, original datetime columns, and `trip_duration` (target) were dropped from the feature set X. Categorical features (`weather`, `store_and_fwd_flag`, `vendor_id`) were one-hot encoded.
- The data was split into training (80%) and testing (20%) sets using `train_test_split`.

- **MLflow Integration:** MLflow was used to track experiments for both models, logging parameters, metrics (MSE, R-squared), and model artifacts.
- **Model 1: Linear Regression**
 - Trained a LinearRegression model on the cleaned training data.
 - Evaluated its performance on the cleaned test data, calculating MSE and R-squared.
 - Logged the model and its metrics to MLflow.
- **Model 2: Gradient Boosting Regressor**
 - Trained a GradientBoostingRegressor with default hyperparameters (`n_estimators=100`, `learning_rate=0.1`) on the cleaned training data.
 - Evaluated its performance on the test data, calculating MSE and R-squared.
 - Logged the model and its metrics to MLflow.

Results:

- **Linear Regression:** MSE = 9,516,939.78, R-squared = 0.03
- **Gradient Boosting Regressor:** MSE = 9,505,976.93, R-squared = 0.03

Conclusion: The Gradient Boosting Regressor showed a marginally better R-squared score, indicating slightly superior performance in explaining the variance in trip duration, albeit both models performed poorly with the given feature set.

5. Data Drift Simulation and Monitoring

Process:

- **Data Drift Simulation:** A 'rush hour surge' scenario was simulated on a copy of the original DataFrame (`df_drift`). This involved:
 - Increasing passenger counts during defined rush hour periods (7-9 AM, 4-6 PM) for a percentage of trips.
 - Slightly increasing `distance_km` for rush hour trips (simulating detours/traffic).
 - Increasing `trip_duration` for rush hour trips (simulating congestion).
- The `df_drift` was preprocessed and one-hot encoded to match the model's expected input features.

- **Performance Monitoring:** The trained Linear Regression model (`lr_model`) was used to make predictions on the `X_drift` data.
- MSE and R-squared were calculated for the model's performance on the drifted data.
- **Retraining Trigger:** A performance degradation threshold was defined (10% drop in R-squared from original performance).
- If the R-squared on drifted data fell below this threshold, a retraining process was simulated (retraining the `LinearRegression` model on the original `X_train_cleaned` and `y_train_cleaned`).
- The retrained model's performance was evaluated and logged to MLflow.

Results:

- R-squared on Drifted Data (Linear Regression): 0.0170
- Original R-squared (Linear Regression): 0.0279
- The R-squared value on drifted data fell below the defined threshold (0.0251), triggering a simulated retraining. The retrained model achieved an R-squared of 0.03, demonstrating recovery to original performance.

6. Improving Model Performance: Hyperparameter Tuning

Process:

- Recognizing the low R-squared scores, `GridSearchCV` was initiated for hyperparameter tuning of the `GradientBoostingRegressor`.
- A `param_grid` was defined with variations for `n_estimators`, `learning_rate`, and `max_depth`.
- `GridSearchCV` was configured with 3-fold cross-validation and `r2` as the scoring metric.
- **(Note: The actual `grid_search.fit` and subsequent steps were commented out in the notebook, indicating this was a demonstration of the *approach* rather than a fully executed tuning due to computational constraints for a mini-project.)**
- The intention was to log the best parameters and the performance of the tuned model to MLflow.

7. Model Packaging and Serving

Process:

- **Model Loading:** The best-performing model (Linear Regression based on the ADR's decision due to being the focus of subsequent steps, despite GBR's marginal edge) was loaded from MLflow using its run_id and model_uri.
- **Sample Prediction:** The loaded model was used to make predictions on a sample of test data to demonstrate its functionality.
- **REST API Serving (MLflow mlflow models serve):** Instructions were provided to serve the model locally as a REST API using the mlflow models serve command, requiring the MLflow run ID and specifying a port.
- **Dockerfile Generation (MLflow with Tracking Server):** A requirements.txt and Dockerfile were generated. This Dockerfile was designed to build an image that would fetch the model from an MLflow tracking server at runtime.
- **Dockerfile Generation (Self-Contained):** An alternative, self-contained Dockerfile was generated. This involved:
 - Downloading the model artifacts locally.
 - Copying these artifacts into the Docker image.
 - Modifying the Dockerfile to serve the model directly from the embedded artifacts, removing the dependency on an external MLflow tracking server at runtime.
- Detailed instructions for building and running both types of Docker images were provided.
- A sample JSON payload for testing the model endpoint was also provided.\

8. MLflow UI Access

Process:

- MLflow UI was started in the background on port 5001.

9. Resources

This project utilized several open-source libraries and platforms:

- **pandas:** For data manipulation and analysis.

- **numpy:** For numerical operations.
- **scikit-learn:** For machine learning model implementations (Linear Regression, Gradient Boosting Regressor) and utility functions (train_test_split, GridSearchCV).
- **matplotlib & seaborn:** For data visualization.
- **MLflow:** For experiment tracking, model logging, model registry, and model serving utilities.
- **Docker:** For containerization of the ML model for deployment.

10. Artifacts:

Git Hub location: <https://github.com/2025paml563-coder/mlminiproject>

Versioned data: <https://github.com/2025paml563-coder/mlminiproject/blob/main/data.zip>

Feature engineered data: https://github.com/2025paml563-coder/mlminiproject/blob/main/data_feature_engineered.zip